RELATIONAL ALGEBRA

1. What is the total revenue generated by each customer, and how many orders have they placed?

Γ CustomerID, FirstName, LastName, COUNT(OrderID) → total_orders, SUM(price) → total_revenue

(

  π CustomerID, FirstName, LastName, OrderID, price

  (

    (Customers ⋈ Customers.CustomerID = Orders.CustomerID)

    ⋈ Orders.OrderID = orderDetails.OrderID

)

)

2. Which product categories contribute the most to revenue?

Γ category_name, SUM(price * StockQuantity) → total_revenue

(

  π category_name, price, StockQuantity

  (

    (Category ⋈ Category.CategoryID = Products.CategoryID)

    ⋈ Products.ProductID = OrderDetails.ProductID

)

)

3. What is the average delivery delay per shipping provider?

Γ shipping_provider_name, AVG(DATEDIFF(EstimatedDeliveryDate, ActualDeliveryDate)) → avg_delivery_delay

(

  π shipping_provider_name, EstimatedDeliveryDate, ActualDeliveryDate

  (

    ShippingProviders ⋈ ShippingProviders.ProviderID = delivery.ProviderID

)

)

4. Which customers have the highest number of declined orders?

Γ CustomerID, FirstName, LastName, COUNT(OrderID) → declined_orders

(

  π CustomerID, FirstName, LastName, OrderID

  (

    (σ Status = 'Declined' (Orders)) ⋈ Orders.CustomerID = Customers.CustomerID

)

)

5. What are the top 5 products by total sales revenue?

TOP 5 (

  Γ product_name, SUM(StockQuantity * price) → total_revenue

  (

    π product_name, StockQuantity, price

    (

      Products ⋈ Products.ProductID = OrderDetails.ProductID

)

)

ORDER BY total_revenue DESC

)

6. What is the average order value by customer?

Γ CustomerID, FirstName, LastName, AVG(order_total) → avg_order_value

(

Γ CustomerID, FirstName, LastName, SUM(price) → order_total

(

π CustomerID, FirstName, LastName, price

(

(Customers ⋈ Customers.CustomerID = Orders.CustomerID)

⋈ Orders.OrderID = orderDetails.OrderID

)

)

)

7. Which customers made the most recent orders?

Γ CustomerID, FirstName, LastName, MAX(OrderDate) → most_recent_order

(

π CustomerID, FirstName, LastName, OrderDate

(

Customers ⋈ Customers.CustomerID = Orders.CustomerID

)

)

8. Which products have the lowest stock levels?

LIMIT 10 (

π product_name, StockQuantity

(

Products

)

ORDER BY StockQuantity ASC

)

9. Which product categories have the highest number of orders?

Γ category_name, COUNT(OrderID) → total_orders

(

π category_name, OrderID

(

(Category ⋈ Category.CategoryID = Products.CategoryID)

⋈ Products.ProductID = OrderDetails.ProductID

)

)

10. What is the total revenue per month?

Γ order_year, order_month, SUM(price) → total_revenue

(

π order_year, order_month, price

(

(Orders ⋈ Orders.OrderID = orderDetails.OrderID)

)

)